SOHAN GADEWAR

Pune, Maharashtra, India

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Education

Vishwakarma Institute of Information Technology **2023 – 2027**

B.Tech in Information Technology | ***CGPA: 8.97/10.00*** *Pune, India*

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, System Design, Object Oriented Programming, Machine Learning.

Experience & Leadership

India Innovates Hackathon **Jan 2026 – Feb 2026**

National Finalist & Lead Builder – Ministry of Education & AICTE *Pune, India*

- Ranked among **top 20% finalist teams** nationwide out of **6,000+ participating teams** across India, pitching scalable public service automation.
- Spearheaded product discovery and rapid prototyping, coordinating cross-functional engineering execution under tight deadline constraints.

South African Patent Office **2025**

Patent Applicant & Inventor – AI-Powered Lost & Found Matching System *Patent Filed*

- Conceptualized and filed patent for an AI matching system utilizing a **5-factor blind multimodal scoring engine** (image embeddings, NLP text, geolocation, timestamp) with automated urgency detection.
- Qualified for Round 2 in **GHCI 25 National GenAI Hackathon** (AnitaB.org India & Backbase | 'Unbound with GenAI' | [**Certificate**]), demonstrating production feasibility.

Projects

Project Sentinel: AI Decision Memory & State-Drift Agent | *Python, PostgreSQL, FastAPI, Lovable, Notion* **Aug 2026 – Sept 2026**

- Spearheaded product discovery across engineering standups, Slack, and Jira, identifying critical context loss and finding that unprompted bot summaries cause heavy distraction by interrupting developer focus.
- Formulated the **"Silent-by-Default"** product thesis, passively ingesting audio transcripts and commit streams without channel spam while deliberately scoping out employee surveillance features in V1.
- Architected deterministic audit engine powered by **append-only PostgreSQL schemas**, surfacing discrepancies between spoken commitments and recorded codebase state strictly upon explicit user queries.
- Shipped clickable MVP on Lovable ([**Live Demo**]); authored comprehensive **Notion PRD** detailing state transition matrices, query grammar, and phased enterprise rollout.

CausalRetention: Prescriptive Uplift & Customer Retention Engine | *Python, XGBoost, Streamlit, Docker* **Oct 2026**

- Uncovered key economic friction in retention marketing where conventional churn models incur a **-13.1% net ROI** due to deadweight spend on non-responsive "Lost Causes" and churn acceleration from nudging dormant "Sleeping Dogs".
- Architected prescriptive AI decision engine using a **Two-Model (T-Learner) XGBoost pipeline** to calculate Conditional Average Treatment Effect (CATE), segmenting customer cohorts into 4 distinct behavioral quadrants.
- Engineered a constrained **Knapsack financial budget optimizer** under a fixed $15,000 constraint, mathematically allocating incentives to top incremental-value cohorts and capturing **+$112.7K+ in net preserved revenue (+752% campaign ROI)**.
- Shipped containerized **interactive Streamlit suite** with Docker and automated executive reporting ([**Live Demo & Repo**]); validated incrementality via Qini curves and cumulative uplift against randomized control baselines.

FlashDrop: High-Concurrency Flash Release & ACID Inventory Engine | *Java, Spring Boot, MySQL* **Oct 2026**

- Architected a high-concurrency flash release inventory engine in Spring Boot backed by MySQL 8.0, eliminating race conditions and guaranteeing **zero overselling** via InnoDB pessimistic row-level locking (**SELECT ... FOR UPDATE**).
- Designed a **3NF relational schema with ACID stored procedures** (sp_reserve_product, sp_checkout_order), ensuring atomic multi-table state transitions across limited inventory, buyer holds, and finalized orders.
- Implemented defensive database triggers to physically prevent negative inventory anomalies and automated native **MySQL EVENT SCHEDULER** routines to reclaim expired 5-minute temporary cart reservations.
- Configured optimized **HikariCP connection pooling** with programmatic TransactionTemplate boundaries; verified via a multi-threaded Python stress suite simulating concurrent buyer races with **0% data inconsistencies** ([**GitHub Repo**]).

Technical Skills

Languages: Python, Java, SQL, Golang, C++, C, JavaScript

Frameworks: Spring Boot, FastAPI, Streamlit, XGBoost, Scikit-learn, React

Databases & Cloud: PostgreSQL, MySQL, Supabase, Docker, Git, Linux

Product & Tools: PRD Writing, Jobs-to-be-Done (JTBD), RICE, Product Metrics, Figma, Postman, Notion, LeetCode (400+ solved, 210+ GFG)

Certifications

- **Product Management: Product Management Masterclass (GeeksforGeeks)** — Product lifecycle, JTBD discovery, and roadmap metrics.
- **AI & Workflows:** Prompt Engineering | Advanced RAG (CampusX) | Docker for ML (CampusX) | n8n Automation (CampusX)